Tutorial 6: Worked solutions

Electrical Engineering I · Module 92315 · Tutorial 6

From Fundamentals to AI Apocalypse · personal study guide

Contents

Task 10: Stationary electric flow field **3**

Answers **4**

Made by Thanate Amornjiraporn (Focus)

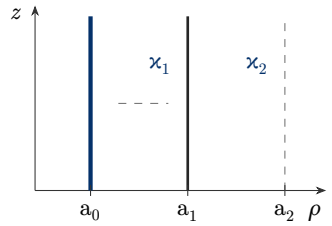
Track the conserved current through each cylindrical sector, use local Ohm's law in each material, then integrate the radial field across both layers.

Made by Thanate Amornjiraporn (Focus)

Task 10: Stationary electric flow field

The conductor is an annular sector with axial height h and angular span α. An ideal inner electrode at radius a_0 feeds current through material $\varkappa_1$ from a_0 to a_1, then through material $\varkappa_2$ from a_1 to a_2 to the outer electrode. Take positive I and U_{12} in the outward direction from electrode 1 to electrode 2, as drawn; use α in radians.

Connected topic: Current density (§2.1.2) and local Ohm's law (§2.2.2) determine the radial fields; resistance (§2.2.4) follows by integrating through the layered path.



Meridian view of the radial current path. The cross-section is revolved through angle α; at radius ρ, the current passes through area $\alpha\,\rho\,h$. The interface at a_1 separates the two conductivities.

10a: Set up the current-density field

Try first: use charge conservation at steady state and the normal-current boundary condition at the $\varkappa_1/\varkappa_2$ interface.

Answers: see "Task 10: Stationary electric flow field" under Answers at the end.

10b: Find J as a function of I

Try first: calculate the area normal to radial flow at radius ρ, then divide the total current by that area.

Answers: see "Task 10: Stationary electric flow field" under Answers at the end.

10c: Find the electric field strength

Try first: apply $J = \varkappa E$ separately in each radial layer.

Answers: see "Task 10: Stationary electric flow field" under Answers at the end.

10d: Find $U_{12}(I)$

Try first: integrate the radial field along the current path, changing conductivity at the material boundary.

Answers: see "Task 10: Stationary electric flow field" under Answers at the end.

10e: Find the resistance

Try first: divide the voltage from 10d by the current, or add the two layer resistances in series.

Answers: see "Task 10: Stationary electric flow field" under Answers at the end.

Made by Thalie Amornjrapon (Focus)

Answers

Task 10: Stationary electric flow field

10a · Continuity across the layers

1. In a stationary flow, charge does not accumulate in the conductor, so the net current through each coaxial sector surface is the same. At the interface $\varrho = a_1$, the normal is e_ϱ: its normal current component is continuous, and the tangential electric-field components (zero here) are continuous too.
2. The current flows radially, normal to the cylindrical interface. Therefore $J_{\varrho,1} = J_{\varrho,2}$. Local Ohm's law $J = \varkappa E$ means the electric field generally changes at the interface:

$$E_{\rho,2}/E_{\rho,1} = \kappa_1/\kappa_2$$

3. The ideal electrodes have $\varkappa \to \infty$, so their internal electric field is zero while they remain equipotential. Neglect edge spreading at the ends of the sector; the current in the two finite-conductivity layers is radial.

Task 10: Stationary electric flow field

10b · Current through a cylindrical sector

1. At radius ϱ, the curved surface spans arc length $\alpha\,\varrho$ and axial height h, so its area normal to the flow is:

$$A(\rho) = \alpha\,\rho\,h$$

2. By symmetry the radial current density is uniform over this surface, and $I = \int J \cdot \mathrm{d}A = J_\varrho A(\rho)$:

$$J(\rho) = [I/(\alpha\,h\,\rho)]e_\rho, \qquad a_0 < \rho < a_2$$

The same expression holds on both sides of a_1. It falls as $1/\varrho$ because the current-carrying area grows with radius.

Task 10: Stationary electric flow field

10c · Piecewise Ohm's law

1. In the first layer, $a_0 < \rho < a_1$, divide the current density by $\varkappa_1$:

$$E_1(\rho) = [I/(\alpha\, h\, \kappa_1 \rho)] e_\rho$$

2. In the second layer, $a_1 < \rho < a_2$, use $\varkappa_2$ instead:

$$E_2(\rho) = [I/(\alpha\, h\, \kappa_2 \rho)] e_\rho$$

At the interface the normal current remains continuous, while E jumps by the conductivity ratio found in 10a. The direction is outward for positive I.

Task 10: Stationary electric flow field

10d · Voltage drop across both layers

1. With U_{12} defined as electrode-1 potential minus electrode-2 potential, it equals the outward field integral from a_0 to a_2. Split the integral where $\varkappa$ changes:

$$U_{12} = \int_{a_0}^{a_1} E_{1,\rho}\, \mathrm{d}\rho + \int_{a_1}^{a_2} E_{2,\rho}\, \mathrm{d}\rho$$

2. Substitute the fields and integrate $\mathrm{d}\rho/\rho$ in each interval:

$$U_{12}(I) = [I/(\alpha\, h\, \kappa_1)] \ln(a_1/a_0) + [I/(\alpha\, h\, \kappa_2)] \ln(a_2/a_1)$$

The voltage is positive for outward positive current. Each logarithm is dimensionless.

Task 10: Stationary electric flow field

10e · Series resistance of radial layers

1. Resistance is $R = U_{12}/I$. Since the current passes through $\varkappa_1$ and $\varkappa_2$ sequentially, their radial resistances add:

$$R_1 = \ln(a_1/a_0)/(\alpha\, h\, \kappa_1), \qquad R_2 = \ln(a_2/a_1)/(\alpha\, h\, \kappa_2)$$

2. Therefore:

$$R = 1/(\alpha\, h)[\ln(a_1/a_0)/\kappa_1 + \ln(a_2/a_1)/\kappa_2]$$

The units are ohms: $1/(\varkappa h)$ has units Ω, and the angle and logarithms are dimensionless. If $\varkappa_1 = \varkappa_2 = \varkappa$, the logs combine to $\ln(a_2/a_0)$, a useful check.

Personal worked-solution guide based on the supplied EE1 Tutorial Task sheet, written in my own words; no tutorial text or figures are reproduced. Not an official FAU publication; check course details against current material. Made by **Thanate Amornjiraporn (Focus)**.

Made by Thanate Amornjiraporn (Focus)